TRUSTJET TRUSTSCORE — WHERE THE DATA STANDS
Plain-language status report — 9 November 2026

WHAT THIS DOCUMENT IS

A non-technical companion to backend/docs/TRUSTSCORE_DATA_SOURCES.md. It answers three questions: what the TrustScore can tell a broker today, what it cannot yet see, and which of those gaps need a decision from the business rather than more engineering.

Every number here was read from the live database on 9 November 2026.

1. THE SHORT VERSION

The scoring algorithm is complete and running. All 1,839 operators have a score. Six public data sources feed it, and four of the six risk categories a broker sees are now backed by real evidence.

Two categories cannot be fully assessed, and in both cases the obstacle is access to an FAA system rather than unfinished work on our side. We have confirmed this from two independent networks, so it is not a configuration problem we can engineer around.

One further input — the age of an operator's charter certificate — is an acknowledged estimate. Both routes to a real figure have been tested and neither works.

Nothing in the system currently overstates an operator. Where we cannot see something, the report says so.

2. WHAT THE SCORE IS BUILT FROM TODAY

Source	What it tells us	Status
FAA accident records (NTSB)	Crashes and serious incidents	Live
FAA enforcement actions	Regulatory penalties against the operator	Live
US DOT legal orders	Consumer-protection and contract findings	Live
FAA mechanical reports (SDRS)	Component failures and emergency procedures	Live
FAA Part 135 roster (weekly)	Which aircraft an operator is legally allowed to fly for hire	Live
FAA aircraft registry (daily)	Loans and liens recorded against a specific aircraft	Live

Currently held in the database

Operators scored	1,839
Aircraft tracked	11,167

Aircraft legally authorised for charter	11,188
Charter certificates covered	1,663
Accident records	247
DOT enforcement orders	677
Aircraft with an active lien	175
Trading names resolved to a certificate	1,914

3. ONE THING WORTH KNOWING: TRADING NAMES

An operator that rebrands used to lose its entire history. Accident and enforcement records are filed under whatever name the company used at the time, so searching only the current name returned nothing and the operator scored as though it had a clean record.

We now resolve every operator to all 1,914 names its certificate has traded under, in both directions. Searching finds the history regardless of renaming.

This was not a theoretical risk. One operator scored 98.93 out of 100 – the top tier – purely because nothing matched its current name.

4. WHAT THE SCORE CANNOT YET SEE

4.1 AIRCRAFT OWNERSHIP AND TITLE – needs FAA access

A broker wants to know whether an aircraft has clean title: no outstanding loans, no mechanic's liens, no history of being shuffled between shell companies.

We can see part of this. The FAA publishes a rolling index of documents filed in roughly the last 180 days, and we read it – 99 active loans or liens across 175 aircraft.

What we cannot see is anything older. A lien recorded three years ago and never released simply does not appear in that index. So finding nothing proves nothing, and the report correctly shows this category as "not assessed" rather than "clear".

The permanent record exists, in an FAA system called CARES. It holds every bill of sale, loan and lien release for an aircraft since it was built.

WE CANNOT REACH IT. The system refuses our requests, from the production server and from a developer machine on a different network. This is not a firewall or routing issue we can work around – the FAA appears to require registered access.

>> DECISION NEEDED: obtain FAA registered access to CARES, or accept that title verification stays incomplete. This is a procurement question, not an engineering one.

4.2 FLEET STABILITY – needs time, and the weekly job to keep running

Rapid fleet turnover is a recognised warning sign: an operator shedding or absorbing aircraft quickly may be in trouble. Measuring it means comparing today's fleet against the same fleet twelve months ago.

The measurement is built and working. What we lack is history. We began recording fleet composition recently and currently hold 31 days.

We cannot backfill this. The FAA overwrites its weekly file, and the

Internet Archive – which we checked, expecting a few years of snapshots – holds exactly ONE historical copy. So the only way to get this data is to keep collecting it.

Usable signal: around MAY 2027.
Full twelve-month measure: around NOVEMBER 2027.

>> ACTION: a weekly automated job must run without interruption. Every week missed pushes those dates back, and the data cannot be recovered later.

4.3 CERTIFICATE AGE – an acknowledged estimate

How long an operator has held its charter certificate affects its score: a thirty-year-old operation is treated differently from a six-month-old one.

We do not have this date. The FAA's published roster does not include it.

Two substitutes were tested and both failed:

- Business registration date. Available for about half of operators, and it measures something different – when the company was formed, not when it was certified to fly charters.

- Aircraft registration dates from the FAA registry. Looked promising at 98% coverage, but it agreed with the other source only 16% of the time, with disagreements running equally in both directions. The reason is structural: that date resets whenever an aircraft is sold, so it measures recent purchases, not operator history.

The one system that holds the real date, FAA AVInfo, is unreachable for the same reason as CARES above.

Current behaviour: where the date is unknown we assume ten years and FLAG THE SCORE AS BASED ON AN ESTIMATE. The assumption is close to the real median we measured (10.7 years), so it is reasonable on average – but it is not evidence for any individual operator.

>> DECISION NEEDED: request certificate issue dates directly from the FAA office that publishes the roster, or accept the flagged estimate. We recommend accepting it; the uncertainty is disclosed, not hidden.

4.4 TWO SMALLER GAPS

Minor safety occurrences. Runway excursions, gear problems and bird strikes that never reach a formal accident report are not currently collected. A public database exists; this is straightforward work, not blocked.

Industry watchdog reports. Commercial services publish operator complaint and dispute data that would strengthen the commercial-integrity assessment.

>> DECISION NEEDED: whether to license watchdog data. This is a cost and terms-of-service question.

5. HOW WE HANDLE WHAT WE CANNOT SEE

Each operator report carries six categories, colour-coded:

GREEN	checked, nothing adverse found
YELLOW	something found, but not severe or not recent
RED	serious and current
GREY	NOT ASSESSED – we could not check this properly

The grey state matters more than it sounds. Without it, "we found nothing" and "we could not look" produce the same clean-looking report, and the second is misleading. Any category resting on a source we could not query is marked grey rather than green.

Two deliberate consequences:

- Aircraft title stays grey even when we DO find liens, because the index we read cannot prove the absence of older ones.
- Fleet stability stays grey until enough history accumulates.

We also corrected a related problem. An operator with a 1988 accident was showing RED on flight safety — reading to a broker as a current hazard — while the score itself had correctly discounted a 38-year-old event to almost nothing. Flags now age at the same rate as the score. The accident is still disclosed in the written report; it is no longer presented as current.

6. WHAT WE RECOMMEND

Now, no decision required

1. Re-score all 1,839 operators. Current scores predate the lien data and several corrections, so they are out of date. Roughly two days of unattended processing.
2. Confirm the weekly FAA collection jobs are scheduled and monitored. This is the only gap that gets worse if ignored.

Needs a business decision

3. FAA registered access for CARES and AVInfo. Would complete title verification and remove the certificate-age estimate. Procurement, not engineering — no amount of development substitutes for it.
4. Industry watchdog data licensing. Cost and terms question.

Straightforward engineering, not blocked

5. Minor safety occurrence collection.

7. THE HONEST SUMMARY

The TrustScore is built on six live public data sources and produces a defensible score for every operator in the database. Four of six risk categories are evidenced.

The two that are not are limited by access to FAA systems, which we have verified from two networks. We have not worked around this and do not believe it can be worked around with development effort.

Where the system cannot see something it says so, rather than implying a clean result. That is the single most important property to preserve: a score that overstates an operator is worse than one that admits uncertainty.

Technical detail, measurements and source references:
backend/docs/TRUSTSCORE_DATA_SOURCES.md

REGENERATING THE PDF

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cd backend/docs
cupsfilter -o cpi=12 -o lpi=7 -o page-left=54 -o page-right=54 \
           -o page-top=54 -o page-bottom=54 \
           TRUSTSCORE_STATUS.txt > TRUSTSCORE_STATUS.pdf
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cupsfilter ships with macOS. The project venv has no PDF library and pip installs are unavailable in this environment, which is why this is a fixed-width text document rather than a designed one. Keep lines at or under 80 characters or they will wrap awkwardly at 12 cpi.